Principles and Techniques of Modern Radar Systems

1. The term Radar cross section defines the:
a. Power radiating ability of the radar
b. Amount of energy scattered by unwanted objects
c. Cross section of radar area through which energy is emitted
d. Scattering ability of the target

Correct answer: d. Scattering ability of the target

2. Which of the following assumption does not hold true for the derivation of RADAR equation?
a. Medium is free space.
b. Presence of scatterers other than target.
c. Target is at situated at far field from the transmitting antenna.
d. Radar system is lossless.

Correct answer: b. Presence of scatterers other than target.

3. A monostatic RADAR operating at X band has a receiver sensitivity of 20 dBm. If the sensitivity is increased by 15 percent. What is the percentage change in the maximum radar range?
a. Decreased by 3.43 percent.
b. Increased by 3.43 percent.
c. Decreased by 6.86 percent.
d. Increased by 6.86 percent.

Correct answer: a. Decreased by 3.43 percent.

4. In a pulsed radar system if the echo due to the first pulse arrives after sending next n pulses, this echo is called:
a. nth time around echo
b. (n-1) th time around echo
c. (n+1) th time around echo
d. n(n+1) th time around echo

Correct Answer: c. (n+1) th time around echo

5. The radar range equation is applicable for:
a. Continuous wave radar
b. Pulsed radar
c. Frequency modulated continuous wave radar
d. All of these

Correct Answer: d. All of these

6. Which of the following statements accurately describes the relationship between antenna beamwidth and antenna aperture in terms of wavelength?
a. Antenna beamwidth decreases as the aperture size decreases
b.Antenna beamwidth increases as the aperture size decreases
c.Antenna beamwidth increases as the aperture size increases
d.Antenna beamwidth is independent of the aperture size

Correct Answer: b. Antenna beamwidth increases as the aperture size decreases

7. If the peak transmitted power in a radar system is increased by a factor of 81 the maximum radar range will be:
a. Increased by a factor of 9.
b. Decreased by a factor of 9.
c. Increased by a factor of 3.
d. Decreased by a factor of 3.

Correct Answer: c. Increased by a factor of 3

8. Range beyond which targets appear as second time around echoes is called
a. Minimum unambiguous range.
b. Maximum unambiguous range.
c. Maximum RADAR range.
d. Minimum RADAR range.

Correct answer: b. Maximum unambiguous range.

9. Which parameter primarily affects the range resolution of a radar system?
a. Frequency
b. Pulse width
c. Antenna size
d. Receiver sensitivity

Correct answer: b. Pulse width

10. In Rayleigh region, RCS of a sphere is
a. Inversely proportional to the wavelength.
b. Directly proportional to square of the wavelength.
c. Inversely proportional to the fourth power of wavelength.
d. Inversely proportional to the square of the wavelength.

Correct answer: c. Inversely proportional to the fourth power of wavelength.

11. During the RCS measurement of a spherical object, which of the following facility is not needed?
a. Facility of changing polarization.
b. Facility of object rotation.
c. Facility of changing frequency.
d. All the above-mentioned facilities are needed.

Correct answer: b. Facility of object rotation.

12.RCS measured using a scaled model is reliable when
a. The target is made of metal.
b. The target is made of a homogeneous dielectric material.
c. The target is made of a composite material having finite conductivity, permittivity and permeability.
d. Any of the above.

Correct answer: a. The target is made of metal

13. What is the receiver sensitivity (in dBm)?
a. -74
b.-87
c.-95
d.-106
Correct answer: d. -106

14. What is the average transmitter power to achieve a maximum unambiguous range of 50 Km?
a.500 kw
b.500 MW
c.1 KW
d.1 MW

Correct answer: a. 500 kw

15. Consider the following statements:
A. In MTI RADAR range measurement is unambiguous and it is used for low speed target.
B. In PULSED DOPPLER RADAR velocity measurement is unambiguous and range measurement is ambiguous.

a. Statement A is correct.
b. Statement B is correct.
c. Both statements are correct.
d. Both the statements are incorrect.

Correct answer: c. Both statements are correct.

16. Determine the frequency at which the RADAR is operating.
a. 5 MHz
b. 7.5 MHz
c. 12.5 MHz
d. 15 MHz

Correct answer: b. $7.5\ MHz$

17. Determine the distance at which the target is positioned away from the RADAR.
a. 30 Km
b. 60 Km
c. 90 Km
d. 100 Km

Correct answer: b. $60\ Km$

18. Beat frequency is a measure of
a. Range of the target.
b. Velocity of the target.
c. Both range and velocity of the target.
d. None of the above.

Correct answer: c. Both range and velocity of the target.

19. Determine the maximum unambiguous range that an S band MTI radar operating at 2.4 GHz if the first blind speed occurs at 480 km/h.
a. 35 km
b. 70 km
c. 125 km
d. 250 km

Correct Answer: b. $70\ km$

20. Echo from a stationary target is called
a. Doppler Shift.
b. Clutter
c. Beats
d. None of these

Correct answer: b. Clutter.

21. The desired MTI filter frequency response should not have
a.A null at f=0
b.A peak at f=0
c.Either a null or a peak at f=0
d.Neither a null nor a peak at f = 0

Correct answer: b. A peak at f=0

22. The concept of feedback in DLCs is conceived to
a. Get a sharp notch in the filter response
b. Make the passband of the filter more uniform
c. Both (a) and (b)
d. None of the above

Correct answer: c. Both (a) and (b)

23. For a single DLC MTI filter, the ratio of clutter attenuation to MTI improvement factor is 1:8. What is the ratio of pulse repetition frequency to clutter spread?
a. √6 π:1
b. 4π : 1
c. 1:√6 π
d. 1: 4π

Correct answer: b. 4π:1

24. The technique to determine the location of the target by measuring the difference in the echo signals received from a moving target is called
a. Sequential lobing
b. Conical scan
c. Simultaneous lobing
d. Squint beam technique

Correct answer: a. Sequential lobing

25. A radar measures an apparent range of 45 km when the PRF is 4000 Hz, but it measures an apparent range of 52.5 km when the PRF is 2500 Hz. What is the true range of the target?
a. 105 km
b. 232.5 km
c. 365.75 km
d. 465 km

Correct answer: b. 232.5 km

26. The squinted beam is advantageous over a direct beam in tracking radar because
a. squinted beam gives a stronger error signal
b. squinted beam gives a weaker error signal
c. squinted beam gives an error signal which is more sensitive to change
d. none of the above

Correct Answer: c. squinted beam gives an error signal which is more sensitive to change

27. In the context of airborne radar systems, which factor poses a significant challenge to MTI filtering, impacting its effectiveness in distinguishing desired targets from clutter?
a. Limiting Doppler resolution due to high carrier frequency
b. Insufficient clutter rejection capabilities
c. Challenges in accurately extracting range information
d. Interference from blind speeds with the desired target Doppler space

Correct answer: d. Interference from blind speeds with the desired target Doppler space

28. What is the primary difference between search radars and tracking radars?
a. Tracking radars have wider beam widths compared to tracking radars.
b. Tracking radars employ electronic scanning antennas, while search radars use mechanical scanning antennas.
c. Search radars have higher angular resolution compared to search radars.
d. Tracking radars use pencil beams for azimuth and elevation measurements, while search radars use fan beams.

Correct answer: d. Tracking radars use pencil beams for azimuth and elevation measurements, while search radars use fan beams.

29. Error in angle measurement in case of sequential lobing and conical scan RADAR occurs due to which of the following reasons?
a. The target RCS is fluctuating.
b. Angle determination time is equal to the time required to receive at least four echo pulses.
c. If a mechanical scan antenna is used then keeping alignment with the reference is difficult.
d. All of these.

Correct answer: d. All of these.

30. What is the typical minimum angular accuracy of an instrumentation radar system?
a. Equal to the beamwidth of the antenna
b. 1/20 of the beamwidth
c. 1/100 of the beamwidth
d. 1/200 of the beamwidth

Correct answer: d. $1/200$ of the beam width

31. A magic tee is a microwave junction which works on the principle of superposition of
a. power
b. energy
c. voltage
d. none of these

Correct Answer: c. voltage

32. Determine the delay at the output of the matched filter, if the round-trip delay is 2.5 $msec$.
a. 0.5 $msec$
b. 1 $msec$
c. 2.5 $msec$
d. 5 $msec$

Correct answer: d. 5 $msec$

33. Considering a top wall coupler acting as a MONOPULSE comparator, if the target is at the MONOPULSE axis then,
a. Diagonal difference is zero.
b. Sum is equal to two times the voltage measured at the feed horn.
c. Total input power is equal to the total output power.
d. All of the above.

Correct answer: d. All of the above.

34.3 dB locus of the combined range and Doppler ambiguity function is
a. A parabola
b. An ellipse
c. A hyperbola
d. A circle

Correct answer: b. An ellipse

35.Consider the following statements:
I. The minimum separation between two targets which the radar can distinguish as separate targets is referred to as the range resolution of the radar.
II. For a transmitted rectangular pulse, the range ambiguity function is a triangular pulse. III. Autocorrelation function is an even function.

a. Only statement I is correct.
b. Only statement III is correct.
c. Both statement I and II are correct.
d. All the statements are correct.

Correct answer: d. All the statements are correct

36. For a rectangular pulse, if the delay resolution is multiplied by a factor of k, it is observed that the range resolution is improved by a factor of 4. Determine k.
a. 8
b. 4
c. 1
d. 0.25

Correct answer: d. 0.25

37. Range ambiguity function should have the following properties:
a. Sharp peak at τ=0
b. High roll-off around τ=0
c. Either (a) or (b)
d. Both (a) and (b)

Correct answer: d. Both (a) and (b)

38. What does range resolution in radar refer to?
a. The maximum distance a radar system can detect a target.
b. The ability of a radar system to accurately measure the velocity of a target.
c. The minimum separation required between two targets for the radar to distinguish them as separate targets.
d. The signal-to-noise ratio (SNR) of a radar system.

Correct answer: c. The minimum separation required between two targets for the radar to distinguish them as separate targets.

39. The product of the delay resolution time constant and Doppler resolution constant is always constant for
a. chirp signal
b. a single rectangular pulse
c. a coherent train of rectangular pulses
d. both a single rectangular pulse and a train of coherent pulses

Correct Answer: d. both a single rectangular pulse and a train of coherent pulses

40. In a radar, using an upchirp and rectangular pulse, the operating frequency sweeps from 4.97 GHz to 5 GHz within 15 µs. Determine the range compression gain of the LFM pulse.
a. 450
b. 400
c. 350
d. 300

Correct answer: a. 450